

PRONROY DAS

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PhD candidate in Electrical and Computer Engg. at Purdue spanning machine learning + optimization for physical systems, quantum optics and quantum theory. Proficiency in programming (Python/PyTorch, C++, CUDA) and strong publication record. Seeking a Student Researcher Intern position.

Education

Purdue University

Aug 2021 – Dec 2027 (Expected)

Ph.D. in Electrical & Computer Engineering (ECE), GPA: 3.89/4.0

West Lafayette, IN

- **Selected Awards:** Hugh and Edna Donnan Fellowship (2026), APS Distinguished Student Award (2025), Purdue ECE Bravo Award (2024)
- Authored 10 journal articles (3 first authorships) ([list of publications](#))

Indian Institute of Science Education and Research (IISER) Kolkata

Aug 2016 – Jun 2021

Dual BS-MS in Physics, GPA: 8.82/10

West Bengal, India

Relevant Experience

ML + Optics Researcher

May 2026 – Present

Center for Integrated Nanotechnologies (CINT), Sandia National Laboratories

Albuquerque, NM

- (Ongoing) Building an amortized sampler for Boltzmann machines via correlated diffusion, realized on a continuous-variable optical Ising system engineered for fully frustrated (spin-glass) couplings, a physics-based approach to hard sampling and optimization problems.

HPC Research Consultant

Mar 2026 – May 2026

Zetascope LLC

West Lafayette, IN

- Developed a low-power, field-deployable faint-object detection pipeline running in near real-time on quantum shot-noise-limited images.
- Engineered optimized CUDA C++ kernel on NVIDIA Blackwell architecture, cutting runtime by $\sim 108\times$.

Graduate Research Assistant (Ph.D.)

Aug 2021 – Present

Prof. Zubin Jacob's Group, Birck Nanotechnology Center, Dept. of ECE, Purdue University

West Lafayette, IN

- Revealed the key quantum mechanism behind the strong dependence between rotation of polarization of light and its wavelength as it transmits through a special class of materials (*Phys. Rev. B (Letter)*, *Editor's Suggestion*).
- Developed an electromagnetic simulation package (C++) that predicts how polarized light interacts with semiconductors using the formalism developed above, reaching $5\times$ lower error compared to existing methods.
- Established the correct quantum theory for the angular momentum (twist) of structured light, predicting a new, measurable fundamental quantum noise (*New J. Phys.*; *JOSA B*).
- Co-designed a cryogenic confocal microscope for single spin-qubit (diamond defect) control and readout; created qubit defects via ion implantation; built all-optical qubit control system ([multiple articles](#)).

Undergraduate Researcher

Aug 2019 – Jul 2021

Prof. Chiranjib Mitra's Group, Dept. of Physics, IISER Kolkata

West Bengal, India

- Designed (CST) and fabricated (photolithography) on-chip microwave antennas for <10 K spin spectroscopy (*J. Phys. D: Appl. Phys.*); built a lock-in confocal microscope to optically manipulate diamond spin qubits.

Skills

- **Programming:** Python (NumPy, SciPy, PyTorch), C++, CUDA, Matlab; simulation in CST Studio and COMSOL
- **Machine learning & optimization:** generative modeling (diffusion models, energy-based models / Boltzmann machines), Ising and combinatorial optimization
- **Quantum theory:** Quantum optics, quantum field theory, quantum information, condensed matter physics

Outreach and Leadership

- Scientific reviewer for APS Publishing, IOP Publishing, The Royal Society Publishing, and SciPost.
- Graduate course instructor for 60+ students with a team of 6 TAs; awarded for excellence in teaching.
- Multimedia and Web Design Team Lead: Inquivesta 2018- Led team for India's largest science fest